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Integrated circuit having latch with transistors of different gate widths

Abstract

An integrated circuit includes a first inverter and a first transmission gate constructed with wide type-one transistors and wide type-two transistors. The integrated circuit also includes a first clocked inverter constructed with narrow type-one transistors and narrow type-two transistors. A latch is formed with the first inverter and the first clocked inverter. The first transmission gate is connected to between an output of the first inverter. The wide type-one transistors are formed in a wide type-one active-region structure and the narrow type-one transistors are formed in a narrow type-one active-region structure. The wide type-two transistors are formed in a wide type-two active-region structure and the narrow type-two transistors are formed in in a narrow type-two active-region structure.

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Background/Summary

PRIORITY CLAIM (1) The present application is a continuation of U.S. application Ser. No. 18/160,630, filed Jan. 27, 2023, now U.S. Pat. No. 12,003,242, issued Jun. 4, 2024, which claims the priority of U.S. Provisional Application No. 63/381,887, filed Nov. 1, 2022, each of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) The recent trend in miniaturizing integrated circuits (ICs) has resulted in smaller devices which

consume less power yet provide more functionality at higher speeds. The miniaturization process has also resulted in stricter design and manufacturing specifications as well as reliability challenges. Various electronic design automation (EDA) tools generate, optimize and verify standard cell layout designs for integrated circuits while ensuring that the standard cell layout design and manufacturing specifications are met.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a circuit diagram of an integrated circuit having a master-slave flip-flop and the associated supporting circuits, in accordance with some embodiments.
- (3) FIGS. 2A-2G are floor plans of the integrated circuit of FIG. 1, in accordance with some embodiments.
- (4) FIGS. 3A-3D are circuit diagrams of various variations of the integrated circuit in FIG. 1, in accordance with some embodiments.
- (5) FIG. 4A is a circuit diagram of an integrated circuit having a data latch and a clock supporting circuit, in accordance with some embodiments.
- (6) FIG. 4B is a floor plan of the integrated circuit of FIG. 4A, in accordance with some embodiments.
- (7) FIGS. 5A-5E are floor plans of a multi-cell integrate circuit that includes multiple single-height circuit cells and a multiple-height circuit cell having the integrated circuit of FIG. 1, in accordance with some embodiments.
- (8) FIG. 6 is a circuit diagram of flip-flop circuits in using with a combination logic circuit, in accordance with some embodiments.
- (9) FIG. 7 is a flowchart of a method of fabricating an integrated circuit, in accordance with some embodiments.
- (10) FIG. 8 is a block diagram of an electronic design automation (EDA) system in accordance with some embodiments.
- (11) FIG. 9 is a block diagram of an integrated circuit (IC) manufacturing system, and an IC manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship

to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(14) In some embodiments, a master-slave flip-flop includes a transmission gate coupled between a master latch and a slave latch. The master latch includes a first inverter and a first clocked inverter. The slave latch includes a second inverter and a second clocked inverter. The output of the first inverter and the input of the first clocked inverter are connected to the input of the transmission gate, while the input of the first inverter and the output of the first clocked inverter are connected together. The input of the second inverter and the output of the second clocked inverter are connected to the output of the transmission gate, while the output of the second inverter and the input of the second clocked inverter are connected together. The first inverter, the transmission gate, and the second inverter are constructed with wide transistors in wide active-region structures, whereby the time delay along the forward data path in the master-slave flip-flop is reduced. The first clocked inverter and the second clocked inverter are constructed with narrow transistors in narrow active-region structures. The average gate width of the wide transistors is larger than the average gate width of the narrow transistors. In some embodiments, the first inverter, the transmission gate, and the second inverter in the wide active-region structures also form an ordered list following the forward data path in the master-slave flip-flop circuit, whereby the time delay along the forward data path in the master-slave flip-flop is reduced.

(15) FIG. 1 is a circuit diagram of an integrated circuit **100** having a master-slave flip-flop and the associated supporting circuits, in accordance with some embodiments. The master-slave flip-flop is formed with a clock gated input circuit **101**, a master latch **102**, a transmission gate **104**, a slave latch **106**, and an output driver **109**. The master latch **102** is coupled between the clock gated input circuit **101** and the transmission gate **104**. The slave latch **106** is coupled between the transmission gate **104** and the output driver **109**. The associated supporting circuits includes a clock supporting circuit **105** implemented with two inverters **110A** and **110B**. The associated supporting circuits also includes a data input supporting circuit **107** implemented with an inverter **110C**. The clock supporting circuit **105** receives a clock signal CP and generates an inverted clock signal CPB and an in-phase clock signal CPBB. The data input supporting circuit **107** receives a scan enable signal SE and generates an inverted scan enable signal SEB.

(16) The clock gated input circuit **101** includes a transmission gate **110F1**, a gated inverter **110D**, and a gated inverter **110E**. The gated inverter **110D** is configured to receive a data signal D, and the gated inverter **110E** is configured to receive a scan signal SI. Each of the gated inverter **110D** and the gated inverter **110E** includes a first part (labeled correspondingly as **110Dp** and **110Ep**) having the PMOS transistors and a second part (labeled correspondingly as **110Dn** and **110En**) having the NMOS transistors.

(17) Each of the gated inverter **110D** and the gated inverter **110E** is configured to receive the scan enable signal SE and the inverted scan enable signal SEB. The combination of the gated inverter **110D** and the gated inverter **110E** forms a multiplexer. When the scan enable signal SE is at logic LOW (and the inverted scan enable signal SEB is at logic HIGH), the data signal D is selected as the input signal coupled to the transmission gate **110F1**, and the scan signal SI is decoupled from the transmission gate **110F1**. Alternatively, when the scan enable signal SE is at logic HIGH (and the inverted scan enable signal SEB is at logic LOW), the scan signal SI is selected as the input signal coupled to the transmission gate **110F1**, and the data signal D is decoupled from the transmission gate **110F1**.

(18) The transmission gate **110F1** is configured to receive the inverted clock signal CPB and the in-phase clock signal CPBB. When the inverted clock signal CPB is at logic LOW (and the in-phase clock signal CPBB is at logic HIGH), none of the data signal D and the scan signal SI is

transmitted at the input node ml_ax of the master latch **102**. When the inverted clock signal CPB is at logic HIGH (and the in-phase clock signal CPBB is at logic LOW), the inverse of the selected input signal (which is either the data signal D or the scan signal SI) is transmitted to the input node ml_ax of the master latch **102**. Here, the inverse of the data signal D is transmitted to the input node ml_ax when the scan enable signal SE is at logic LOW, but the inverse of the scan signal SI is transmitted to the input node ml_ax when the scan enable signal SE is at logic HIGH.

(19) In FIG. **1**, the master latch **102** includes an inverter **110G** and a clocked inverter **110F2** driven by the clock signals CPB and CPBB. The output of the inverter **110G** is connected to the input of the clocked inverter **110F2**, while the output of the clocked inverter **110F2** is connected to the input of the inverter **110G**. When the clock signal CPB is at logic HIGH (and the clock signal CPBB is at logic LOW), the master latch **102** is at the unlatched state, the signal at the output node ml_b of the master latch **102** is the inverse of the signal at the input node ml_ax of the master latch **102**.

Consequently, the signal at the output node ml_b becomes the inverse of the selected input signal (which is either the data signal D or the scan signal SI) received from the clock gated input circuit **101**. Meanwhile, the transmission gate **104** is set to the non-connected state by the clock signals CPB and CPBB, and the input node sl_a of the slave latch **106** is decoupled from the output node ml_b of the master latch **102**. Thereafter, when the inverted clock signal CPB is change to logic LOW (and the in-phase clock signal CPBB is change to logic HIGH), the master latch **102** is changed to the latched state, the signal at the output node ml_b of the master latch **102** is maintained during the time period that the clock signal CPB at logic LOW (and the clock signal CPB at logic HIGH). The latched signal value at the output node ml_b is the signal value of the selected input signal (received from the clock gated input circuit **101**) at a first falling edge of the clock signal CPB, as the clock signal CPB is changed to logic LOW.

(20) When the clock signal CPB is change to logic LOW (and the clock signal CPBB is change to logic HIGH), the transmission gate **104** is changed to the connected state by the clock signals CPB and CPBB, and the latched signal at the output node ml_b of the master latch **102** is transmitted to the input node sl_a of the slave latch **106**. In the embodiment as shown in FIG. **1**, the transmission gate **104** is implemented as a transmission gate **110H1** which has a PMOS transistor receiving the clock signal CPB as the gate voltage and has a NMOS transistor receiving the clock signal CPBB as the gate voltage.

(21) In FIG. **1**, the slave latch **106** includes an inverter **110I** and a clocked inverter **110H2** driven by the clock signals CPB and CPBB. The output of the inverter **110I** is connected to the input of the clocked inverter **110H2**, while the output of the clocked inverter **110H2** is connected to the input of the inverter **110I**. When the clock signal CPB is at logic LOW (and the signal CPBB is at logic HIGH), the slave latch **106** is at the unlatched state, and the signal at the output node sl_bx of the slave latch **106** is the inverse of the signal at the input node sl_a of the slave latch **106**.

Consequently, the inverse of the latched signal at the output node ml_b of the master latch **102** is transmitted to the output node sl_bx of the slave latch **106**. Thereafter, when the clock signal CPB is change to logic HIGH (and the signal CPBB is change to logic LOW), the slave latch **106** is changed to the latched state, the signal at the output node sl_bx of the slave latch **106** is maintained, and the latched signal value at the output node sl_bx after the current rising edge of the clock signal CPB is the inverse of the latched signal at the output node ml_b which is latched after the previous falling edge (the first falling edge) of the clock signal CPB.

(22) During the time period when the clock signal CPB is at logic HIGH after the current rising edge of the clock signal CPB, the signal value $Q_{\text{sub}.n+1}$ at the output Q of the output driver **109** (which is implemented as an inverter **110J**) is equal to the signal value of the selected input signal (received from the clock gated input circuit **101**) at the previous falling edge (the first falling edge) of the clock signal CPB. That is, $Q_{\text{sub}.n+1}=D_{\text{sub}.n}$ when the scan enable signal SE is at logic LOW, and $Q_{\text{sub}.n+1}=SI_{\text{sub}.n}$ when the scan enable signal SE is at logic LOW.

(23) In FIG. **1**, each of the gated inverter **110D**, the transmission gate **110F1**, the inverter **110G**, the

transmission gate **110H1**, the inverter **110I**, and the inverter **110J** is implemented as a class-one device, while each of the clocked inverter **110F2** and the clocked inverter **110H2** is implemented as class-two device. The average gate width of the PMOS transistors in a class-one device is larger than the average gate width of the PMOS transistors in a class-two device. The average gate width of the NMOS transistors in a class-one device is larger than the average gate width of the NMOS transistors in a class-two device. The PMOS transistors in the class-one devices are fabricated in a wide PMOS active-region structure, while the PMOS transistors in the class-two devices are fabricated in a narrow PMOS active-region structure. The NMOS transistors in the class-one devices are fabricated in a wide NMOS active-region structure, while the NMOS transistors in the class-two devices are fabricated in a narrow NMOS active-region structure.

(24) In some embodiments, when the active-region structures are formed with fin structures, the PMOS transistors fabricated in the PMOS active-region structure and in the narrow PMOS active-region structures are p-channel FinFETs, and the NMOS transistors fabricated in the NMOS active-region structure and in the narrow NMOS active-region structures are n-channel FinFETs. In some embodiments, when the active-region structures are formed with nano-sheet structures, the PMOS transistors fabricated in the PMOS active-region structure and in the narrow PMOS active-region structures are p-channel nano-sheet transistors, and the NMOS transistors fabricated in the NMOS active-region structure and in the narrow NMOS active-region structures are n-channel nano-sheet transistors. In some embodiments, when the active-region structures are formed with nano-wire structures, the PMOS transistors fabricated in the PMOS active-region structure and in the narrow PMOS active-region structures are p-channel nano-wire transistors, and the NMOS transistors fabricated in the NMOS active-region structure and in the narrow NMOS active-region structures are n-channel nano-wire transistors.

(25) FIGS. 2A-2G are floor plans of the integrated circuit **100** of FIG. 1, in accordance with some embodiments. In FIGS. 2A-2G, the inverter **110A**, the inverter **110B**, the inverter **110C**, the gated inverter **110D**, and the gated inverter **110E** are implemented correspondingly in layout areas “A”, “B”, “C”, “D”. and “E”. The transmission gate **110F1**, the clocked inverter **110F2**, the inverter **110G**, the transmission gate **110H1**, and the clocked inverter **110H2** are implemented correspondingly in layout areas “F1”, “F2”, “G”, “H1”. and “H2”.

(26) Each of the integrated circuits in FIGS. 2A-2G includes a wide PMOS active-region structure **220P** extending in the X-direction, a wide NMOS active-region structure **220N** extending in the X-direction, a narrow PMOS active-region structure **230P** extending in the X-direction, and a narrow NMOS active-region structure **230N** extending in the X-direction. Each of the gated inverter **110D**, the transmission gate **110F1**, the inverter **110G**, the transmission gate **110H1**, the inverter **110I**, and the inverter **110J** (correspondingly in the layout areas “D”, “F1”, “G”, “H1”, “I”, and “J”) is constructed with wide transistors in the wide PMOS active-region structure **220P** and the wide NMOS active-region structure **220N**. Each of the clocked inverter **110F2** and the clocked inverter **110H2** is implemented (correspondingly in the layout areas “F2” and “H2”) is constructed with narrow transistors in the narrow PMOS active-region structure **230P** and the narrow NMOS active-region structure **230N**.

(27) In FIGS. 2A-2G, each of the inverters **110A**, **110B**, and **110C** is selectively implemented either with wide transistors in the wide active-region structures (i.e., **220P** and **220N**) or with narrow transistors in the narrow active-region structures (i.e., **230P** and **230N**). In the floor plan **200A** of FIG. 2A, the inverter **110A** is implemented with wide transistors in the wide active-region structures (i.e., **220P** and **220N**), while the inverters **110B** and **110C** are implemented with narrow transistors in the narrow active-region structures (i.e., **230P** and **230N**). In the floor plan **200B** of FIG. 2B, the inverter **110B** is implemented with wide transistors in the wide active-region structures (i.e., **220P** and **220N**), while the inverters **110A** and **110C** are implemented with narrow transistors in the narrow active-region structures (i.e., **230P** and **230N**). In each of the floor plans **200C-200G** as shown in FIGS. 2C-2G, the inverter **110C** is implemented with wide transistors in

the wide active-region structures (i.e., **220P** and **220N**), while the inverters **110A** and **110B** are implemented with narrow transistors in the narrow active-region structures (i.e., **230P** and **230N**).

(28) Some example placements of the two wide active-region structures (i.e., **220P** and **220N**) and the two narrow active-region structures (i.e., **230P** and **230N**) are depicted in FIGS. 2A-2G.

(29) In the floor plans **200A-200C** as shown in FIGS. 2A-2C, the two wide active-region structures (i.e., **220P** and **220N**) are laterally positioned between the two narrow active-region structures (i.e., **230P** and **230N**). In the floor plan **200A** of FIG. 2A, the narrow PMOS active-region structure **230P** is adjacent to the wide PMOS active-region structure **220P**, while the narrow NMOS active-region structure **230N** is adjacent to the wide NMOS active-region structure **220N**. In both the floor plan **200B** of FIG. 2B and the floor plan **200C** of FIG. 2C, the narrow PMOS active-region structure **230P** is adjacent to the wide NMOS active-region structure **220N**, while the narrow NMOS active-region structure **230N** is adjacent to the wide PMOS active-region structure **220P**.

(30) In the floor plans **200D-200E** as shown in FIGS. 2D-2E, the two narrow active-region structures (i.e., **230P** and **230N**) are laterally positioned between the two wide active-region structures (i.e., **220P** and **220N**). In the floor plan **200D** of FIG. 2D, the wide PMOS active-region structure **220P** is adjacent to the narrow PMOS active-region structure **230P**, while the wide NMOS active-region structure **220N** is adjacent to the narrow NMOS active-region structure **230N**. In the floor plan **200E** of FIG. 2E, the wide NMOS active-region structure **220N** is adjacent to the narrow PMOS active-region structure **230P**, while the wide PMOS active-region structure **220P** is adjacent to the narrow NMOS active-region structure **230N**.

(31) In the floor plans **200F-200G** as shown in FIGS. 2F-2G, the two wide active-region structures (i.e., **220P** and **220N**) forms a first pair of adjacent active-region structures, and the two narrow active-region structures (i.e., **230P** and **230N**) forms a second pair of adjacent active-region structures. In the floor plan **200F** of FIG. 2F, the wide PMOS active-region structure **220P** is adjacent to the narrow PMOS active-region structure **230P**, while the two PMOS active-region structures (i.e., **220P** and **230P**) are positioned between the wide NMOS active-region structure **220N** and the narrow NMOS active-region structure **230N**. In the floor plan **200G** of FIG. 2G, the wide NMOS active-region structure **220N** is adjacent to the narrow NMOS active-region structure **230N**, while the two NMOS active-region structures (i.e., **220N** and **230N**) are positioned between the wide PMOS active-region structure **220P** and the narrow PMOS active-region structure **230P**.

(32) In FIGS. 2A-2G, the gated inverter **110D**, the transmission gate **110F1**, the inverter **110G**, the transmission gate **110H1**, the inverter **110I**, and the inverter **110J** (which are all class-one devices) are constructed with wide transistors in the wide active-region structures (i.e., **220P** and **220N**). In addition, the gated inverter **110D**, the transmission gate **110F1**, the inverter **110G**, the transmission gate **110H1**, the inverter **110I**, and the inverter **110J** also form an ordered list arranged along the X-direction.

(33) The order of the devices in the ordered list is determined by the direction of the data signal propagation. In the examples of FIGS. 2A-2G, the data signal propagates from the gated inverter **110D** to the transmission gate **110F1**, then from the transmission gate **110F1** to the inverter **110G**, then from the inverter **110G** to the transmission gate **110H1**, then from the transmission gate **110H1** to the inverter **110I**, and then from the inverter **110I** to the inverter **110J**. When the ordered list of the class-one devices is arranged along the X-direction, the X-coordinates of the class-one devices in the ordered list changes monotonously following the order of the class-one devices in the list. For example, in FIGS. 2A-2G, the X-coordinate of the gated inverter **110D** is smaller than the X-coordinate of the transmission gate **110F1**, the X-coordinates of the transmission gate **110F1** is smaller than the X-coordinate of the inverter **110G**, the X-coordinates of the inverter **110G** is smaller than the X-coordinate of the transmission gate **110H1**, the X-coordinates of the transmission gate **110H1** is smaller than the X-coordinate of the inverter **110I**, and the X-coordinates of the inverter **110I** is smaller than the X-coordinate of the inverter **110J**.

(34) In some embodiments, the floor plans of FIGS. 2A-2G are modified to accommodate various

variations of the integrated circuit **100** in FIG. **1**. The clock gated input circuit **101** of FIG. **1** includes a transmission gate **110F1**, a gated inverter **110D**, and a gated inverter **110E**, which forms a clock gated multiplexer. Other implementations of the clock gated input circuit are within the contemplated scope of the present disclosure. As a first example, in the integrated circuit **300A** of FIG. **3A**, the clock gated input circuit **101** is implemented as a gated inverter **110D** serially connected with a transmission gate **110F1**. As a second example, in the integrated circuit **300B** of FIG. **3B**, the clock gated input circuit **101** is implemented as an inverter **310D** serially connected with a transmission gate **110F1**. The inverter **310D** and the transmission gate **110F1** forms a clocked inverter, which is used as the clock gated input circuit **101**.

(35) In some embodiments, the floor plan for the integrated circuit **300A** of FIG. **3A** or the integrated circuit **300B** of FIG. **3A** is obtained by modifying one of the floor plans in FIGS. **2A-2G**. For example, in some embodiments, the modification of a floor plan in FIGS. **2A-2G** includes removing the gated inverter **110E** in the layout area “E.” In some embodiments, after removing the gated inverter **110E** in the layout area “E”, one or more dummy devices are implemented in the layout area “E”. In some alternative embodiments, after removing the gated inverter **110E** in the layout area “E,” one of the inverters **110A**, **110B**, and **110C** which used to be implemented with the wide active-region structures (i.e., **220P** and **220N**) is moved to the layout area “E” and implemented with the narrow active-region structures (i.e., **230P** and **230N**). The modification of the floor plan **200A** in FIG. **2A** includes moving the inverter **110A** to the layout area “E.” The modification of the floor plan **200B** in FIG. **2B** includes moving the inverter **110B** to the layout area “E.” For the integrated circuit **300A** of FIG. **3A**, the modification of one of the floor plans **200C-200G** in FIGS. **2C-2G** includes moving the inverter **110C** to the layout area “E.” For the integrated circuit **300B** of FIG. **3B**, however, the modification includes removing the inverter **110C** from the layout area “C.” Other modifications of the floor plans of FIGS. **2A-2G** for implementing the integrated circuit **300A** or **300B** are also within the contemplated scope of the present disclosure.

(36) Another variation of the integrated circuit **100** in FIG. **1** is the integrated circuit **300C** as shown in FIG. **3C**. In the integrated circuit **300C**, the inverter **110G** in the master latch **102** of FIG. **1** is implemented as a resettable inverter **310G**, and the clocked inverter **110H2** in the slave latch **106** of FIG. **1** is implemented as a resettable clocked inverter **310H2**. A reset logic signal CD is coupled to each of the resettable inverter **310G** and the resettable clocked inverter **310H2**. When the reset logic signal CD is at logic LOW, the resettable inverter **310G** in FIG. **3C** functions the same as the inverter **110G** in FIG. **1**, and the resettable clocked inverter **310H2** in FIG. **3C** functions the same as the clocked inverter **110H2** in FIG. **1**. When the reset logic signal CD is at logic HIGH, however, the signal at the output node ml_b of the master latch **102** is driven to logic LOW, and the slave latch **106** is driven to the unlatched state. In addition, during the time period that the clock signal CPB is at logic LOW, the logic LOW at the output node ml_b passes through the transmission gate **110H1**; then, the logic LOW at the input node sl_a of the slave latch **106** passes through both the inverters **110I** and **110J**. Consequently, the signal at the output Q of the output driver **109** is reset to logic LOW by the reset logic signal CD.

(37) In some embodiments, the floor plan for the integrated circuit **300C** of FIG. **3C** is obtained by modifying one of the floor plans of FIGS. **2A-2G**. For example, in some embodiments, the modification of a floor plan in FIGS. **2A-2G** includes enlarging the layout area “G” along the X-direction to accommodate the newly added wide transistors in the resettable inverter **310G**, and the modification also includes enlarging the layout area “H2” along the X-direction to accommodate the newly added narrow transistors in the resettable clocked inverter **310H2**. Other modifications of the floor plans of FIGS. **2A-2G** for implementing the integrated circuit **300C** are also within the contemplated scope of the present disclosure.

(38) Still another variation of the integrated circuit **100** in FIG. **1** is the integrated circuit **300D** as shown in FIG. **3D**. In the integrated circuit **300D**, the output driver **109** is implemented as an

inverter **310J** having two PMOS transistors and two NMOS transistors, whereby the driving strength of the output driver **109** is increased, as compared with an output driver that is implemented as an inverter having just one PMOS and one NMOS transistor. In some embodiments, to increase the driving strength of the output driver **109** further, more than two PMOS transistors and more than two NMOS transistors are used in the inverter for the output driver **109**. Other variations of the output driver **109** are within the contemplated scope of the present disclosure. For example, in some embodiments, the output driver **109** is implemented as a buffer circuit, and the output signal of the buffer circuit and the input signal of the buffer circuit has the same logic value. In some embodiments, the floor plan for the integrated circuit **300D** of FIG. **3D** is obtained by modifying one of the floor plans of FIGS. **2A-2G**. For example, in some embodiments, the modification of a floor plan in FIGS. **2A-2G** includes enlarging the layout area “J” along the X-direction to accommodate the newly added wide transistors in the inverter **310J**. Other modifications of the floor plans of FIGS. **2A-2G** for implementing the integrated circuit **300D** are also within the contemplated scope of the present disclosure.

(39) In FIG. **1**, the master-slave flip-flop is used as an example. In the master-slave flip-flop, class-one devices are constructed with wide transistors in wide active-region structures, and class-two devices are constructed with narrow transistors in narrow active-region structures. In the embodiment as shown in FIG. **4A**, a data latch is used as another example. In the data latch, class-one devices are constructed with wide transistors in wide active-region structures, and a class-two device is constructed with narrow transistors in narrow active-region structures.

(40) FIG. **4A** is a circuit diagram of an integrated circuit **400** having a data latch and a clock supporting circuit, in accordance with some embodiments. The data latch includes an inverter **410D**, a transmission gate **410F1**, an inverter **410G**, a clocked inverter **410F2**, an inverter **410K1**, and an inverter **410K2**. The clock supporting circuit **105**, implemented with two inverters **110A** and **110B**, receives a clock signal CP and generates an inverted clock signal CPB and an in-phase clock signal CPBB. The inverter **410D** and the transmission gate **410F1** are connected as a clocked inverter, forming a clock gated input circuit **401**. The inverter **410K1** and the inverter **410K2** forms an output driver **409**. Each of the inverter **410G** and the clocked inverter **410F2** is coupled between the clock gated input circuit **401** and the output driver **409**. The output of the inverter **410G** is connected to the input of the clocked inverter **410F2**, while the output of the clocked inverter **410F2** is connected to the input of the inverter **410G**.

(41) Each of the transmission gate **410F1** and the clocked inverter **410F2** is driven by the clock signals CPB and CPBB. When the inverted clock signal CPB is at logic HIGH (and the in-phase clock signal CPBB is at logic LOW), the inverse of the data signal D is transmitted to the input (i.e., the node ml_ax) of the inverter **410G**, and the logic value at the output (i.e., the node ml_b) of the inverter **410G** is the same as the logic value of the data signal D, because the clocked inverter **410F2** is disabled by the clock signals CPB and CPBB. Thereafter, when the inverted clock signal CPB is change to logic LOW (and the in-phase clock signal CPBB is change to logic HIGH), the logic value at the output (i.e., the node ml_b) of the inverter **410G** is latched by the clocked inverter **410F2** because the clocked inverter **410F2** is enabled by the clock signals CPB and CPBB. Furthermore, during the time period that the clock signal CPB at logic LOW (and the in-phase clock signal CPBB is at logic HIGH), the transmission gate **410F1** also prevents the data signal D from transmitting to the input (i.e., the node ml_ax) of the inverter **410G**. The logic value at the output (i.e., the node ml_b) of the inverter **410G** is transmitted to the output of the inverter **410K2** as the latched output signal Q.

(42) FIG. **4B** is a floor plan of the integrated circuit **400** of FIG. **4A**, in accordance with some embodiments. In FIG. **4B**, each of the wide PMOS active-region structure **220P**, the wide NMOS active-region structure **220N**, the narrow PMOS active-region structure **230P**, and the narrow NMOS active-region structure **230N** extends in the X-direction. The inverter **410D**, the transmission gate **410F1**, the inverter **410G**, the inverter **410K1**, and the inverter **410K2** are

implemented correspondingly in layout areas “D”, “F1”, “G”, “K1”, and “K2”, while the inverter **110A**, the clocked inverter **410F2**, the inverter **110B** are implemented correspondingly in layout areas “A”, “F2”, and “B”. Each of the inverter **410D**, the transmission gate **410F1**, the inverter **410G**, the inverter **410K1**, and the inverter **410K2** is implemented as a class-one device which is constructed with wide transistors in the wide active-region structures (i.e., **220P** and **220N**), while the clocked inverter **410F2** is implemented as a class-two device which is constructed with narrow transistors in the narrow active-region structures (i.e., **230P** and **230N**). In FIG. 4B, each of the inverters **110A** and **110B** is also constructed with narrow transistors in the narrow active-region structures (i.e., **230P** and **230N**). In addition, as shown by the order of the layout areas “D”, “F1”, “G”, “K1”, and “K2”, an ordered list of the inverter **410D**, the transmission gate **410F1**, the inverter **410G**, the inverter **410K1**, and the inverter **410K2** is arranged along the X-direction in the floor plan.

(43) The floor plan in FIG. 4B is provided as an example, various variations of the floor plans in FIGS. 2A-2G adapted for the integrated circuit **400** of FIG. 4A are within the contemplated scope of the present disclosure. Other implementations of the floor plan for the integrated circuit **400** of FIG. 4A are also within the contemplated scope of the present disclosure.

(44) In some embodiments, the integrated circuit **100** in FIG. 1 is designed as a circuit cell based on the floor plans in FIGS. 2A-2G. In some embodiments, the integrated circuit **400** of FIG. 4A is designed as a circuit cell based on the floor plan in FIG. 4B. The circuit cell of the integrated circuit **100** or the integrated circuit **400** is often placed in a larger multi-cell integrated circuit that has other single-height circuit cells. In the larger multi-cell integrated circuit, most of the logic gates (such as NOT gates, NOR gates, and NAND gates) are implemented as single-height circuit cells, while the integrated circuit **100** or the integrated circuit **400** is implemented as a double-height circuit cell or a triple-height circuit cell.

(45) FIGS. 5A-5E are floor plans of a multi-cell integrate circuit that includes multiple single-height circuit cells and a multiple-height circuit cell having the integrated circuit **100** of FIG. 1, in accordance with some embodiments. In FIGS. 5A-5E, a circuit cell **500** occupying multiple cell-rows is placed between two cell rows **510** and **520**. The cell row **510** is identified as between horizontal lines **501** and **503**, and the cell row **520** is identified as between horizontal lines **502** and **504**. Along the cell row **510** extending in the X-direction, multiple single-height circuit cells (not all shown in the figure) are aligned between the horizontal lines **501** and **503**, and one of the single-height circuit cells is identified as the single-height circuit **514**. The single-height circuit **514** is implemented in the layout area “U” in each of the floor plans. Along the cell row **520** extending in the X-direction, multiple single-height circuit cells (not all shown in the figure) are aligned between the horizontal lines **502** and **504**, and one of the single-height circuit cells is identified as the single-height circuit **524**. The single-height circuit **524** is implemented in the layout area “V” in each of the floor plans. The cell height of each cell is measured along the Y-direction in the unit of “CH”. The Y-direction is perpendicular to the X-direction.

(46) Each of the single-height circuit **514** and the single-height circuit **524** has a cell height that is equal to $1.0 \times \text{CH}$. The circuit cell **500** (which is implemented with the integrated circuit **100** of FIG. 1) is bounded between the horizontal lines **501** and **502**. In FIG. 5A, the circuit cell **500** has a cell height that is equal to $3.0 \times \text{CH}$. In FIGS. 5B-5E, the circuit cell **500** has a cell height that is equal to $2.0 \times \text{CH}$.

(47) In FIGS. 5A-5E, the circuit cells in the cell row **510** are constructed with PMOS transistors in the PMOS active-region structure **542P** and NMOS transistors in the NMOS active-region structure **542N**. The circuit cells in the cell row **520** are constructed with PMOS transistors in the PMOS active-region structure **544P** and NMOS transistors in the NMOS active-region structure **544N**. The PMOS active-region structures **542P** and **544P** and the NMOS active-region structures **542N** and **544N**, which extend in the X-direction, are all in parallel with the wide active active-region structures (i.e., **220P** and **220N**) and the narrow active active-region structures (i.e., **230P** and

230N).

(48) The circuit cell **500** in FIGS. 5A-5E is implemented with the integrated circuit **100** of FIG. 1. Each floor plan of the circuit cell **500** in FIGS. 5A-5E corresponds to one of the floor plans in FIGS. 2B-2C and 2E-2G. In some embodiments, the NMOS active-region structures (i.e., **220N**, **230N**, **542N** and **544N**) are fabricated in a p-type substrate, and the PMOS active-region structures (i.e., **220P**, **230P**, **542P** and **544P**) are fabricated in n-type wells in the p-type substrate.

(49) In FIG. 5A, the circuit cell **500** has a floor plan that is the same as the floor plan **200B** in FIG. 2B, floor plan **200B** has already been described with reference to FIG. 2B. In FIG. 5A, the PMOS active-region structure **542P** and the narrow PMOS active-region structure **230P** are fabricated in n-type well **550AS**, the wide PMOS active-region structure **220P** is fabricated in n-type wells **550AL**, and the PMOS active-region structure **544P** is fabricated in n-type well **554A**. In some embodiments, the n-type wells **550AS**, **550AL**, and **554A** are equally spaced along the Y-direction.

(50) In FIG. 5B, the circuit cell **500** has a floor plan that is the same as the floor plan **200C** in FIG. 2C, and the floor plan **200C** has already been described with reference to FIG. 2C. In FIG. 5B, the wide PMOS active-region structure **220P** and the narrow PMOS active-region structure **230P** are correspondingly fabricated in n-type wells **550BL** and **550BS**, while the PMOS active-region structures **542P** and **544P** are correspondingly fabricated in n-type wells **552B** and **554B**. In some embodiments, the n-type wells **552B**, **550BS**, **550BL**, and **554B** are equally spaced along the Y-direction.

(51) In FIG. 5C, the circuit cell **500** has a floor plan that is the same as the floor plan **200E** in FIG. 2E, and the floor plan **200E** has already been described with reference to FIG. 2E. In FIG. 5C, the wide PMOS active-region structure **220P** and the narrow PMOS active-region structure **230P** are correspondingly fabricated in n-type wells **550CL** and **550CS**, while the PMOS active-region structures **542P** and **544P** are correspondingly fabricated in n-type wells **552C** and **554C**. In some embodiments, the n-type wells **552C**, **550CS**, **550CL**, and **554C** are equally spaced along the Y-direction.

(52) In FIG. 5D, the circuit cell **500** has a floor plan that is the same as the floor plan **200F** in FIG. 2F, and the floor plan **200F** has already been described with reference to FIG. 2F. In FIG. 5D, the wide PMOS active-region structure **220P** and the narrow PMOS active-region structure **230P** are fabricated in n-type well **550D**, while the PMOS active-region structures **542P** and **544P** are correspondingly fabricated in n-type wells **552D** and **554D**. In some embodiments, the n-type wells **552D**, **550D**, and **554D** are equally spaced along the Y-direction.

(53) In FIG. 5E, the circuit cell **500** has a floor plan that is the same as the floor plan **200G** in FIG. 2G, and the floor plan **200G** has already been described with reference to FIG. 2G. In FIG. 5E, the wide PMOS active-region structure **220P** and the narrow PMOS active-region structure **230P** are correspondingly fabricated in n-type wells **550EL** and **550ES**. In the n-type wells **550EL** and **550ES**, the PMOS active-region structures **542P** and **544P** are also correspondingly fabricated. In some embodiments, the n-type wells **550EL** and **550ES** and the adjacent n-type wells (not shown in the figure) are equally spaced along the Y-direction.

(54) In FIG. 1, the gated inverter **110D**, the transmission gate **110F1**, the inverter **110G**, the transmission gate **110H1**, the inverter **110I**, and the inverter **110J** are all on the forward data path of the flip-flop circuit. In FIGS. 2A-2G, each of the devices on the forward data path is implemented as a class-one device constructed with transistors in the wide active-region structures (i.e., **220P** and **220N**). Additionally, FIGS. 2A-2G, the gated inverter **110D**, the transmission gate **110F1**, the inverter **110G**, the transmission gate **110H1**, the inverter **110I**, and the inverter **110J** are arranged in the floor plan along the X-direction as an ordered list following the forward data path of the flip-flop circuit. When the flip-flop circuit is implemented based on one of the floor plans as shown in FIGS. 2A-2G, the setup slack time at the second transmission gate **H1** of the flip-flop circuit is improved, as compared with some alternative implementations in which some devices on the forward data path are not implemented as class-one devices. When the flip-flop circuit is

implemented based on one of the floor plans as shown in FIGS. 2A-2G, the setup slack time at the second transmission gate H1 of the flip-flop circuit is also improved, as compared with some alternative implementations in which the gated inverter 110D, the transmission gate 110F1, the inverter 110G, the transmission gate 110H1, the inverter 110I, and the inverter 110J are not arranged as an ordered list following the forward data path.

(55) When the flip-flop circuits are used with a combination logic circuit, the setup slack time is related to the time delay of the gated inverter 110D, the transmission gate 110F1, the inverter 110G, the transmission gate 110H1, the inverter 110I, and the inverter 110J on the forward data path. FIG. 6 is a circuit diagram of flip-flop circuits in using with a combination logic circuit, in accordance with some embodiments. The circuit in FIG. 6 includes flip-flop circuits 620 and 640 and a combination logic circuit 630 coupled between the flip-flop circuits 620 and 640. Each of the flip-flop circuits 620 and 640 is implemented as an instance of the flip-flop circuit in FIG. 1, and each of the flip-flop circuits 620 and 640 receives a clock signal CP from a clock input node 611. In FIG. 6, the data signal received at the input D of the flip-flop circuit 620 at the instant of a triggering edge of the clock signal CP is transmitted to the output Q of the flip-flop circuit 620. Subsequently, the data signal at the output Q of the flip-flop circuit 620 passes through the combination logic circuit 630 and arrives at the input D of the flip-flop circuit 640. Then, the data signal at the input D of the flip-flop circuit 640 is transmitted to the transmission gate 110H1 (as shown in FIG. 1) within the flip-flop circuit, which is to be latched by the slave latch 106 (as shown in FIG. 1).

(56) In FIG. 6, the arrival time $T_{sub.arr}$ of the data at the input of the transmission gate 110H1 in the flip-flop circuit 640 is given by the equation

$T_{sub.arr} = T_{sub.clock} + T_{sub.CP2Q} + T_{sub.comb} + T_{sub.D2ml_b}$, where $T_{sub.clock}$ is the time of the triggering edge of the clock signal CP, $T_{sub.CP2Q}$ is the delay time from the triggering input of the flip-flop circuit 620 to the output Q of the flip-flop circuit 620, $T_{sub.comb}$ is the delay time of the combination logic circuit 630, and $T_{sub.D2ml_b}$ is the delay time from the input D of the flip-flop circuit 640 to input of the transmission gate 110H1 in the flip-flop circuit 640. The data required time T_{reg} of the data at the input of the transmission gate 110H1 in the flip-flop circuit 640 is given by the equation $T_{sub.req} = T_{sub.clock} + T_{sub.ck2TXG}$, where $T_{sub.ck2TXG}$ is the delay time from the triggering input of the flip-flop circuit 640 to the input of the transmission gate 110H1 in the flip-flop circuit 640. The setup slack time at the input of the transmission gate 110H1 is the difference between the required time $T_{sub.req}$ and the arrival time $T_{sub.arr}$ of the data at the input of the transmission gate 110H1. When the flip-flop circuits 620 and 640 are implemented with one of the floor plans in FIGS. 2A-2G, each of the delay time $T_{sub.CP2Q}$ and the delay time $T_{sub.D2ml_b}$ decreases, which results in an improvement of the setup slack time at the input of the transmission gate 110H1. Additionally, when the flip-flop circuit 640 is implemented with one of the floor plans in FIGS. 2C-2G, the delay time $T_{sub.ck2TXG}$ also increases, which leads to a further improvement of the setup slack time at the input of the transmission gate 110H1.

(57) FIG. 7 is a flowchart of a method 700 of fabricating an integrated circuit, in accordance with some embodiments. The sequence in which the operations of method 700 are depicted in FIG. 7 is for illustration only; the operations of method 700 are capable of being executed in sequences that differ from that depicted in FIG. 7. It is understood that additional operations may be performed before, during, and/or after the method 700 depicted in FIG. 7, and that some other processes may only be briefly described herein.

(58) In operation 710 of method 700, a wide type-one active-region structure and a narrow type-one active-region structure extending in the X-direction are fabricated. In operation 720 of method 700, a wide type-two active-region structure and a narrow type-two active-region structure extending in the X-direction are fabricated. In the example embodiments as shown in FIGS. 2A-2G and FIGS. 3A-3E, the wide PMOS active-region structure 220P and the narrow PMOS active-region structure 230P are fabricated in operation 710 (or alternatively in operation 720), and the wide NMOS active-region structure 220N and the narrow NMOS active-region structure 230N are

fabricated in operation **720** (or alternatively in operation **710**). In addition, in the example embodiments as shown in FIGS. **3A-3E**, the PMOS active-region structures **542P** and **544P** are fabricated in operation **710** (or alternatively in operation **720**), and the NMOS active-region structures **542N** and **544N** are fabricated in operation **720** (or alternatively in operation **710**). After operations **710** and **720**, the fabrication process proceeds to operation **730** and **740**.

(59) In operation **730** of method **700**, a first inverter, a first transmission gate, and a second inverter are formed with wide type-one transistors in the wide type-one active-region structure and wide type-two transistors in the wide type-two active-region structure. In the example embodiments as shown in FIGS. **2A-2G** and FIGS. **3A-3E**, the inverter **110G**, the transmission gate **110H1**, and the inverter **110I** are formed with wide PMOS transistors in the wide PMOS active-region structure **220P** and wide NMOS transistors in the wide NMOS active-region structure **220N**. In operation **740** of method **700**, a first clocked inverter and a second clocked inverter are formed with narrow type-one transistors in the narrow type-one active-region structure and narrow type-two transistors in the narrow type-two active-region structure. In the example embodiments as shown in FIGS. **2A-2G** and FIGS. **3A-3E**, the clocked inverter **110F2** and the clocked inverter **110H2** are formed with narrow PMOS transistors in the narrow PMOS active-region structure **230P** and narrow NMOS transistors in the narrow NMOS active-region structure **230N**. After operations **710** and **720**, the fabrication process proceeds to operation **750** and **760**.

(60) In operation **750** of method **700**, the first clocked inverter is connected with the first inverter, whereby a first latch circuit is formed. In the example embodiments as shown in FIGS. **2A-2G** and FIGS. **3A-3E**, the output of the inverter **110G** is connected to the input of the clocked inverter **110F2**, and the output of the clocked inverter **110F2** is connected to the input of the inverter **110G**, whereby the master latch **102** is formed. In operation **760** of method **700**, the second clocked inverter is connected with the second inverter, whereby a second latch circuit is formed. In the example embodiments as shown in FIGS. **2A-2G** and FIGS. **3A-3E**, the output of the inverter **110I** is connected to the input of the clocked inverter **110H2**, and the output of the clocked inverter **110H2** is connected to the input of the inverter **110I**, whereby the slave latch **106** is formed.

(61) In some embodiments, at least some portions of the wide active-region structures and the narrow active-region structures fabricated in operation **710** and operation **720** have uniform widths along the Y-direction. For example, in the floor plans in FIGS. **2F-2G**, each of the width of the wide PMOS active-region structure **220P** and the width of the wide NMOS active-region structure **220N** within at least the layout areas D, F1, G, H1, I, J for implementing the class-one devices remains uniformly the same, each of the width of the narrow PMOS active-region structure **230P** and the width of the narrow NMOS active-region structure **230N** within at least the layout areas F2 and H2 for implementing the class-two devices also remains uniformly the same. In FIGS. **2F-2G**, the uniform width (or the average width) of the wide PMOS active-region structure **220P** is large than the uniform width (or the average width) of the narrow PMOS active-region structure **230P**, and the uniform width of the wide NMOS active-region structure **220N** is large than the uniform width (or the average width) of the narrow NMOS active-region structure **230N**.

(62) In some alternative embodiments, some portions of the wide active-region structures and the narrow active-region structures fabricated in operation **710** and operation **720** do not have uniform widths along the Y-direction. In some floor plans in FIGS. **2A-2E**, the widths of the wide PMOS active-region structure **220P** and the widths of the wide NMOS active-region structure **220N** within some of the layout areas D, F1, G, H1, I, J for implementing the class-one devices are individually adjusted to further improve the slack time of the flip flop in FIG. **1**. In some floor plans in FIGS. **2A-2E**, the widths of the narrow PMOS active-region structure **230P** and the widths of the narrow NMOS active-region structure **230N** within some of the layout areas F2 and H2 for implementing the class-two devices are individually adjusted.

(63) As an example, in the embodiments as shown in FIGS. **2B** and **2D**, the widths of the wide PMOS active-region structure **220P** within the layout areas D, F1, G, H1, I, J are not uniformly the

same, and the widths of the widths of the wide NMOS active-region structure **220N** within the layout areas D, F1, G, H1, I, J are not uniformly the same, even though each of the width of the narrow PMOS active-region structure **230P** and the width of the narrow NMOS active-region structure **230N** within the layout areas F2 and H2 for implementing the class-two devices remains uniformly the same. In FIGS. 2B and 2D, the average width of the wide PMOS active-region structure **220P** is still large than the uniform width (or the average width) of the narrow PMOS active-region structure **230P** within the layout areas F2 and H2, and average width of the wide NMOS active-region structure **220N** is still large than the uniform width (or the average width) of the narrow NMOS active-region structure **230N** within the layout areas F2 and H2. Consequently, the average gate width of the wide PMOS transistors in the class-one devices is larger than the average gate width of the narrow PMOS transistors in the class-two devices, the average gate width of the wide NMOS transistors in the class-one devices is larger than the average gate width of the narrow NMOS transistors in the class-two devices.

(64) As an example, in the embodiments as shown in FIGS. 2C and 2E, the widths of the narrow PMOS active-region structure **230P** within the layout areas F2 and H2 are not uniformly the same, and the widths of the narrow NMOS active-region structure **230N** within the layout areas F2 and H2 are not uniformly the same, even though each of the width of the wide PMOS active-region structure **220P** and the width of the wide NMOS active-region structure **220N** within the layout areas D, F1, G, H1, I, J for implementing the class-one devices remains uniformly the same. In FIGS. 2C and 2E, the uniform width (or the average width) of the wide PMOS active-region structure **220P** is still large than the average width of the narrow PMOS active-region structure **230P** within the layout areas F2 and H2, and uniform width (or the average width) of the wide NMOS active-region structure **220N** is still large than the average width of the narrow NMOS active-region structure **230N** within the layout areas F2 and H2. Consequently, the average gate width of the wide PMOS transistors in the class-one devices is larger than the average gate width of the narrow PMOS transistors in the class-two devices, the average gate width of the wide NMOS transistors in the class-one devices is larger than the average gate width of the narrow NMOS transistors in the class-two devices.

(65) As an example, in the embodiments as shown in FIG. 2A, the widths of the wide PMOS active-region structure **220P** within the layout areas D, F1, G, H1, I, J are not uniformly the same, and the widths of the widths of the wide NMOS active-region structure **220N** within the layout areas D, F1, G, H1, I, J are not uniformly the same, the widths of the narrow PMOS active-region structure **230P** within the layout areas F2 and H2 are not uniformly the same, and the widths of the narrow NMOS active-region structure **230N** within the layout areas F2 and H2 are not uniformly the same. In FIG. 2A, the average width of the wide PMOS active-region structure **220P** within the layout areas D, F1, G, H1, I, J for implementing the class-one devices is still large than the average width of the narrow PMOS active-region structure **230P** within the layout areas F2 and H2 for implementing the class-two devices, and the average width of the wide NMOS active-region structure **220N** within the layout areas D, F1, G, H1, I, J for implementing the class-one devices is large than the average width of the narrow NMOS active-region structure **230N** within the layout areas F2 and H2 for implementing the class-two devices. Consequently, the average gate width of the wide PMOS transistors in the class-one devices is larger than the average gate width of the narrow PMOS transistors in the class-two devices, the average gate width of the wide NMOS transistors in the class-one devices is larger than the average gate width of the narrow NMOS transistors in the class-two devices.

(66) FIG. 8 is a block diagram of an electronic design automation (EDA) system **800** in accordance with some embodiments.

(67) In some embodiments, EDA system **800** includes an automatic placement and routing (APR) system. Methods described herein of designing layout diagrams represent wire routing arrangements, in accordance with one or more embodiments, are implementable, for example,

using EDA system **800**, in accordance with some embodiments.

(68) In some embodiments, EDA system **800** is a general purpose computing device including a hardware processor **802** and a non-transitory, computer-readable storage medium **804**. Storage medium **804**, amongst other things, is encoded with, i.e., stores, computer program code **806**, i.e., a set of executable instructions. Execution of instructions **806** by hardware processor **802** represents (at least in part) an EDA tool which implements a portion or all of the methods described herein in accordance with one or more embodiments (hereinafter, the noted processes and/or methods).

(69) Processor **802** is electrically coupled to computer-readable storage medium **804** via a bus **808**. Processor **802** is also electrically coupled to an I/O interface **810** by bus **808**. A network interface **812** is also electrically connected to processor **802** via bus **808**. Network interface **812** is connected to a network **814**, so that processor **802** and computer-readable storage medium **804** are capable of connecting to external elements via network **814**. Processor **802** is configured to execute computer program code **806** encoded in computer-readable storage medium **804** in order to cause system **800** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, processor **802** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

(70) In one or more embodiments, computer-readable storage medium **804** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, computer-readable storage medium **804** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, computer-readable storage medium **804** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

(71) In one or more embodiments, storage medium **804** stores computer program code **806** configured to cause system **800** (where such execution represents (at least in part) the EDA tool) to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **804** also stores information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **804** stores library **807** of standard cells including such standard cells as disclosed herein. In one or more embodiments, storage medium **804** stores one or more layout diagrams **809** corresponding to one or more layouts disclosed herein.

(72) EDA system **800** includes I/O interface **810**. I/O interface **810** is coupled to external circuitry. In one or more embodiments, I/O interface **810** includes a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to processor **802**.

(73) EDA system **800** also includes network interface **812** coupled to processor **802**. Network interface **812** allows system **800** to communicate with network **814**, to which one or more other computer systems are connected. Network interface **812** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of noted processes and/or methods, is implemented in two or more systems **800**.

(74) System **800** is configured to receive information through I/O interface **810**. The information received through I/O interface **810** includes one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor **802**. The information is transferred to processor **802** via bus **808**. EDA system **800** is configured to receive information related to a user interface (UI) through I/O interface **810**. The information is stored in computer-readable medium **804** as UI **842**.

(75) In some embodiments, a portion or all of the noted processes and/or methods is implemented

as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is used by EDA system **800**. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

(76) In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

(77) FIG. **9** is a block diagram of an integrated circuit (IC) manufacturing system **900**, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some embodiments, based on a layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using manufacturing system **900**.

(78) In FIG. **9**, IC manufacturing system **900** includes entities, such as a design house **920**, a mask house **930**, and an IC manufacturer/fabricator (fab) **950**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **960**. The entities in system **900** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **920**, mask house **930**, and IC fab **950** is owned by a single larger company. In some embodiments, two or more of design house **920**, mask house **930**, and IC fab **950** coexist in a common facility and use common resources.

(79) Design house (or design team) **920** generates an IC design layout diagram **922**. IC design layout diagram **922** includes various geometrical patterns designed for an IC device **960**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **960** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **922** includes various IC features, such as an active region, gate electrode, source and drain, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **920** implements a proper design procedure to form IC design layout diagram **922**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **922** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **922** can be expressed in a GDSII file format or DFII file format.

(80) Mask house **930** includes data preparation **932** and mask fabrication **944**. Mask house **930** uses IC design layout diagram **922** to manufacture one or more masks **945** to be used for fabricating the various layers of IC device **960** according to IC design layout diagram **922**. Mask house **930** performs mask data preparation **932**, where IC design layout diagram **922** is translated into a representative data file (RDF). Mask data preparation **932** provides the RDF to mask

fabrication **944**. Mask fabrication **944** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **945** or a semiconductor wafer **953**. The design layout diagram **922** is manipulated by mask data preparation **932** to comply with particular characteristics of the mask writer and/or requirements of IC fab **950**. In FIG. 9, mask data preparation **932** and mask fabrication **944** are illustrated as separate elements. In some embodiments, mask data preparation **932** and mask fabrication **944** can be collectively referred to as mask data preparation.

(81) In some embodiments, mask data preparation **932** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **922**. In some embodiments, mask data preparation **932** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

(82) In some embodiments, mask data preparation **932** includes a mask rule checker (MRC) that checks the IC design layout diagram **922** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **922** to compensate for photolithographic implementation effects during mask fabrication **944**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

(83) In some embodiments, mask data preparation **932** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **950** to fabricate IC device **960**. LPC simulates this processing based on IC design layout diagram **922** to create a simulated manufactured device, such as IC device **960**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (DOF), mask error enhancement factor (MEEF), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **922**.

(84) It should be understood that the above description of mask data preparation **932** has been simplified for the purposes of clarity. In some embodiments, data preparation **932** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **922** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **922** during data preparation **932** may be executed in a variety of different orders.

(85) After mask data preparation **932** and during mask fabrication **944**, a mask **945** or a group of masks **945** are fabricated based on the modified IC design layout diagram **922**. In some embodiments, mask fabrication **944** includes performing one or more lithographic exposures based on IC design layout diagram **922**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **945** based on the modified IC design layout diagram **922**. Mask **945** can be formed in various technologies. In some embodiments, mask **945** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **945** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the

opaque regions of the binary mask. In another example, mask **945** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **945**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **944** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **953**, in an etching process to form various etching regions in semiconductor wafer **953**, and/or in other suitable processes.

(86) IC fab **950** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **950** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

(87) IC fab **950** includes fabrication tools **952** configured to execute various manufacturing operations on semiconductor wafer **953** such that IC device **960** is fabricated in accordance with the mask(s), e.g., mask **945**. In various embodiments, fabrication tools **952** include one or more of a wafer stepper, an ion implanter, a photoresist coater, a process chamber, e.g., a CVD chamber or LPCVD furnace, a CMP system, a plasma etch system, a wafer cleaning system, or other manufacturing equipment capable of performing one or more suitable manufacturing processes as discussed herein.

(88) IC fab **950** uses mask(s) **945** fabricated by mask house **930** to fabricate IC device **960**. Thus, IC fab **950** at least indirectly uses IC design layout diagram **922** to fabricate IC device **960**. In some embodiments, semiconductor wafer **953** is fabricated by IC fab **950** using mask(s) **945** to form IC device **960**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **922**. Semiconductor wafer **953** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **953** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

(89) An aspect of the present disclosure relates to an integrated circuit. The integrated circuit includes a wide type-one active-region structure and a wide type-two active-region structure; a narrow type-one active-region structure and a narrow type-two active-region structure. The circuit also includes wide type-one transistors in the wide type-one active-region structure and narrow type-one transistors in the narrow type-one active-region structure, where an average gate width of the wide type-one transistors is larger than an average gate width of the narrow type-one transistors. The circuit also includes wide type-two transistors in the wide type-two active-region structure and narrow type-two transistors in the narrow type-two active-region structure, where an average gate width of the wide type-two transistors is larger than an average gate width of the narrow type-two transistors; a first inverter and a first transmission gate constructed with the wide type-one transistors and the wide type-two transistors, where the first transmission gate is connected to an output of the first inverter; and a first clocked inverter constructed with the narrow type-one transistors and the narrow type-two transistors, where the first clocked inverter has an input connected to the output of the first inverter and has an output connected to an input of the first inverter.

(90) Another aspect of the present disclosure relates to an integrated circuit. The integrated circuit includes a wide type-one active-region structure and a wide type-two active-region structure; a narrow type-one active-region structure and a narrow type-two active-region structure. The circuit also includes wide type-one transistors in the wide type-one active-region structure and narrow type-one transistors in the narrow type-one active-region structure, where an average gate width of

the wide type-one transistors is larger than an average gate width of the narrow type-one transistors. The circuit also includes wide type-two transistors in the wide type-two active-region structure and narrow type-two transistors in the narrow type-two active-region structure, where an average gate width of the wide type-two transistors is larger than an average gate width of the narrow type-two transistors; a gated inverter a first inverter, and an output driver constructed with the wide type-one transistors and the wide type-two transistors, where the first inverter has an input connected to an output of the gated inverter and has an output connected to an input of the output driver; and a first clocked inverter constructed with the narrow type-one transistors and the narrow type-two transistors, where the first clocked inverter has an input connected to the output of the first inverter and has an output connected to the input of the first inverter.

(91) An aspect of the present disclosure relates to a method. The method includes fabricating a wide type-one active-region structure and a narrow type-one active-region structure where an average width of the wide type-one active-region structure is larger than an average width of the narrow type-one active-region structure; fabricating a wide type-two active-region structure and a narrow type-two active-region structure where an average width of the wide type-two active-region structure is larger than an average width of the narrow type-two active-region structure; forming a first inverter and a first transmission gate with wide type-one transistors in the wide type-one active-region structure and wide type-two transistors in the wide type-two active-region structure, where the first transmission gate is connected to an output of the first inverter; forming a first clocked inverter with narrow type-one transistors in the narrow type-one active-region structure and narrow type-two transistors in the narrow type-two active-region structure. The method also includes connecting an input of the first clocked inverter with the output of the first inverter, and connecting the output of the first clocked inverter with an input of the first inverter.

(92) It will be readily seen by one of ordinary skill in the art that one or more of the disclosed embodiments fulfill one or more of the advantages set forth above. After reading the foregoing specification, one of ordinary skill will be able to affect various changes, substitutions of equivalents and various other embodiments as broadly disclosed herein. It is therefore intended that the protection granted hereon be limited only by the definition contained in the appended claims and equivalents thereof.

Claims

1. An integrated circuit comprising: a wide type-one active-region structure and a wide type-two active-region structure; a narrow type-one active-region structure and a narrow type-two active-region structure; wide type-one transistors in the wide type-one active-region structure and narrow type-one transistors in the narrow type-one active-region structure, wherein an average gate width of the wide type-one transistors is larger than an average gate width of the narrow type-one transistors; wide type-two transistors in the wide type-two active-region structure and narrow type-two transistors in the narrow type-two active-region structure, wherein an average gate width of the wide type-two transistors is larger than an average gate width of the narrow type-two transistors; a first inverter and a first transmission gate constructed with the wide type-one transistors and the wide type-two transistors, wherein the first transmission gate is connected to an output of the first inverter; and a first clocked inverter constructed with the narrow type-one transistors and the narrow type-two transistors, wherein the first clocked inverter has an input connected to the output of the first inverter and has an output connected to an input of the first inverter.
2. The integrated circuit of claim 1, further comprising: a gated inverter constructed with the wide type-one transistors and the wide type-two transistors, wherein the first inverter is coupled between an output of the gated inverter and the input of the first transmission gate.
3. The integrated circuit of claim 1, further comprising: a clock gated input circuit having a second transmission gate constructed with the wide type-one transistors and the wide type-two transistors,

wherein the first inverter is coupled between an output of the second transmission gate and the input of the first transmission gate.

4. The integrated circuit of claim 1, further comprising: a clock gated input circuit constructed with the wide type-one transistors and the wide type-two transistors, wherein the first inverter has the input connected to an output of the clock gated input circuit and has the output connected to the input of the first transmission gate, and wherein the clock gated input circuit, the first inverter, the first transmission gate, are arranged as an ordered list.

5. The integrated circuit of claim 1, further comprising: an output driver constructed with the wide type-one transistors and the wide type-two transistors; and a second inverter coupled between an output of the first transmission gate and an input of the output driver.

6. The integrated circuit of claim 5, wherein an ordered list of the first inverter, the first transmission gate, the second inverter, and the output driver are arranged along a first direction.

7. The integrated circuit of claim 1, further comprising: a gated inverter constructed with the wide type-one transistors and the wide type-two transistors, wherein an ordered list of the gated inverter, the first inverter, and the first transmission gate are arranged along a first direction.

8. The integrated circuit of claim 1, wherein the wide type-one active-region structure and the wide type-two active-region structure are laterally positioned between the narrow type-one active-region structure and the narrow type-two active-region structure.

9. The integrated circuit of claim 1, wherein the narrow type-one active-region structure and the narrow type-two active-region structure are laterally positioned between the wide type-one active-region structure and the wide type-two active-region structure.

10. The integrated circuit of claim 1, wherein the wide type-one active-region structure and the wide type-two active-region structure forms a first pair of adjacent active-region structures, and the narrow type-one active-region structure and the narrow type-two active-region structure forms a second pair of adjacent active-region structures.

11. The integrated circuit of claim 10, wherein the wide type-one active-region structure and the narrow type-one active-region structure are laterally positioned between the wide type-two active-region structure and the narrow type-two active-region structure.

12. The integrated circuit of claim 1, wherein a first average width of the wide type-one active-region structure within areas used to construct the wide type-one transistors is larger than a second average width of the narrow type-one active-region structure within areas used to construct the narrow type-one transistors.

13. The integrated circuit of claim 1, wherein a first average width of the wide type-two active-region structure within areas used to construct the wide type-two transistors is larger than a second average width of the narrow type-two active-region structure within areas used to construct the narrow type-two transistors.

14. The integrated circuit of claim 1, wherein the first inverter is a resettable inverter.

15. An integrated circuit comprising: a wide type-one active-region structure and a wide type-two active-region structure; a narrow type-one active-region structure and a narrow type-two active-region structure; wide type-one transistors in the wide type-one active-region structure and narrow type-one transistors in the narrow type-one active-region structure, wherein an average gate width of the wide type-one transistors is larger than an average gate width of the narrow type-one transistors; wide type-two transistors in the wide type-two active-region structure and narrow type-two transistors in the narrow type-two active-region structure, wherein an average gate width of the wide type-two transistors is larger than an average gate width of the narrow type-two transistors; a gated inverter, a first inverter, and an output driver constructed with the wide type-one transistors and the wide type-two transistors, wherein the first inverter has an input connected to an output of the gated inverter and has an output connected to an input of the output driver; and a first clocked inverter constructed with the narrow type-one transistors and the narrow type-two transistors, wherein the first clocked inverter has an input connected to the output of the first inverter and has

an output connected to the input of the first inverter.

16. The integrated circuit of claim 15, the gated inverter is a clocked inverter.

17. The integrated circuit of claim 15, wherein the output driver is either a buffer or an inverter.

18. The integrated circuit of claim 15, wherein an average width of the wide type-one active-region structure within areas used to construct the wide type-one transistors is larger than an average width of the narrow type-one active-region structure within areas used to construct the narrow type-one transistors, and wherein an average width of the wide type-two active-region structure within areas used to construct the wide type-two transistors is larger than an average width of the narrow type-two active-region structure within areas used to construct the narrow type-two transistors.

19. A method comprising: fabricating a wide type-one active-region structure and a narrow type-one active-region structure, wherein an average width of the wide type-one active-region structure is larger than an average width of the narrow type-one active-region structure; fabricating a wide type-two active-region structure and a narrow type-two active-region structure, wherein an average width of the wide type-two active-region structure is larger than an average width of the narrow type-two active-region structure; forming a first inverter and a first transmission gate with wide type-one transistors in the wide type-one active-region structure and wide type-two transistors in the wide type-two active-region structure, wherein the first transmission gate is connected to an output of the first inverter; forming a first clocked inverter with narrow type-one transistors in the narrow type-one active-region structure and narrow type-two transistors in the narrow type-two active-region structure; and connecting an input of the first clocked inverter with the output of the first inverter, and connecting the output of the first clocked inverter with an input of the first inverter.

20. The method of claim 19, comprising: forming a gated inverter with the wide type-one transistors and the wide type-two transistors, wherein an ordered list of the gated inverter, the first inverter, and the first transmission gate are arranged along a first direction.
